

The model is the leak.

Pseudonymisation removes the name from the record. We proved that the trained model still gives the patient away.

Team **unlisted**

6,000

patients, three times our Stage 2 cohort

3 runs

every model trained and attacked three times

0.6%

accuracy we pay for a formal privacy guarantee

Two hospitals. One model. Zero shared records.

The data cannot travel

Federal Law No. 2 of 2019.

UAE health data must not leave the State.

Pooling the records is not an option.

Names are not enough

PDPL Art. 20 asks for pseudonymisation.

It hides the name in the row.

It does not hide the patient in the model.

Presence is the breach

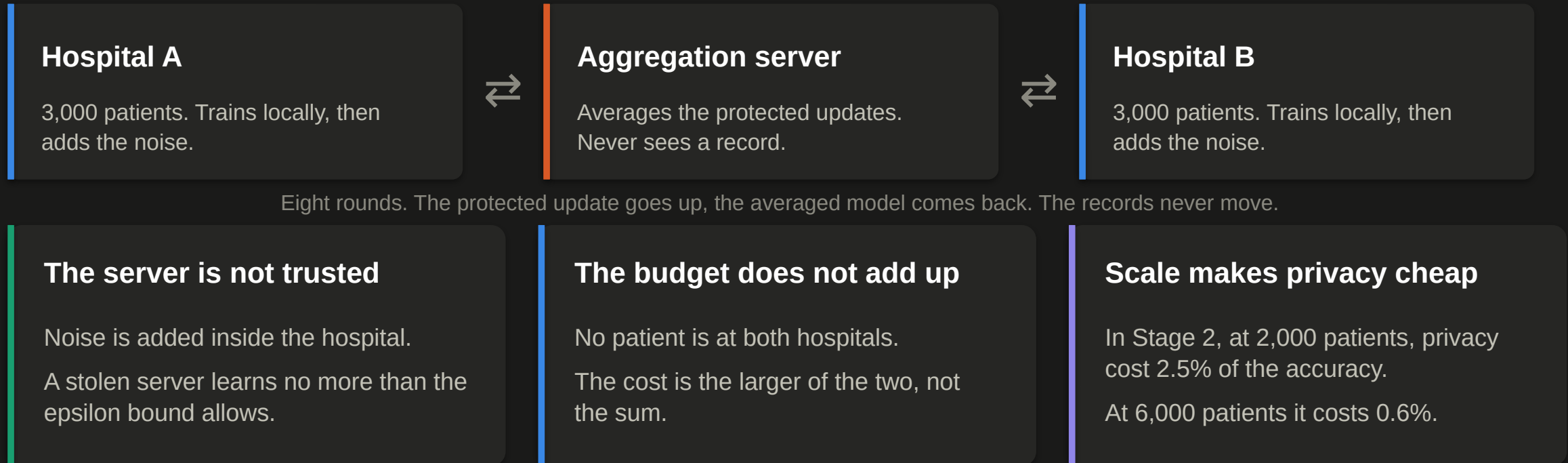
A model that remembers a patient reveals them.

It shows that the person was in that hospital, in that period.

The diagnosis is not needed.

Our job was not to claim this gap exists. It was to measure it.

Records stay home. Only protected maths travels.



More data does not make privacy more expensive. It makes it cheaper.

We attacked our own model with seven attacks, including the state of the art.

0.586

Best of the seven attacks against the careless model. 0.5 means the attacker learns nothing. Our audit proves a real leak at epsilon of 2.36.

0.506

Best of the same seven against the DP model. This is chance level, in every one of the 3 runs.

LiRA, not a simple threshold

96 shadow models for each target.

It learns what a normal score looks like for every single patient.

The average score lies

At a 1% false alarm rate, LiRA finds 1.7 times more patients than the classic attack.

On the average score they look the same.

The leak finds the most ill

Patients with 5 or more conditions score 0.70. Patients with one or two score 0.56.

Only the calibrated attack can see this difference.

A formal guarantee for 0.6% of the accuracy.

MODEL	ACCURACY	ATTACK SCORE	OF THE BEST	GUARANTEE
Trained without care	0.711	0.586	90.7%	none
Trained with care	0.776	0.512	98.9%	none
Federated + DP, epsilon 3	0.771	0.506	98.3%	yes

Budget we spend: epsilon 2.99, delta 0.00001, for each hospital. 0.5 on the attack score is chance level.

Careless is not cheap

The leaking model was also the least accurate model we built.

Privacy and accuracy fail together.

Care is not a guarantee

LiRA still scores 0.512 against the careful model, above chance in all three runs.

Only DP reaches chance level.

Deploy epsilon 3

98.3% of the best possible accuracy, attack held at chance level.

A guarantee that covers future attacks.

The jury asked for scale and repeat runs. We delivered 6,000 patients and 3 independent runs. One command reproduces every number.